

## Kinematics and integration



- 1 The velocity  $v(t)$ , in metres per second, of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$v(t) = 12t, \text{ where } t \geq 0$$

The object is initially at the origin.

- a Find the displacement  $x(t) = \int v(t) dt$  of the particle at time  $t$ .
- b Find the time at which  $x(t) = 54$  m.

- 2 The velocity  $v(t)$ , in metres per second, of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$v(t) = 6t + 10, \text{ where } t \geq 0$$

The object is initially 8 m to the right of the origin.

- a Find the displacement  $s(t)$  of the particle at time  $t$ .
- b Find the displacement of the object after 5 seconds.

- 3 The velocity  $v(t)$ , in metres per second, of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$v(t) = 12t^2 + 30t + 9, \text{ where } t \geq 0$$

The object is initially 6 m to the left of the origin.

- a Find the displacement,  $s(t)$ , of the particle at time  $t$ .
- b Find the displacement of the object after 5 seconds.

- 4 The velocity  $v(t)$ , in metres per second, of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$v(t) = 12\sqrt{t}$$

The object is initially 5 m to the right of the origin.

- a Find the displacement,  $x(t)$ , of the particle at time  $t$ .
- b Hence, find the position of the object after 9 seconds.

- 5 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 30\sqrt{t}$$

- a Find the velocity of the particle after 4 seconds.
  - b Sketch a graph of the velocity of the particle over the first 10 seconds.
  - c Find the area under the velocity function from  $t = 0$  to  $t = 4$ .
  - d Describe what the area found in part (c) represents.
- 6 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 2 - \frac{t}{3}$$

- a Find the velocity of the particle after 9 seconds.
  - b Sketch a graph of the velocity of the particle over the first 12 seconds.
  - c Find the area bounded between the velocity function and the horizontal axis from  $t = 0$  to  $t = 9$ .
  - d Describe what the area found in part (c) represents.
- 7 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 6t + 15$$

The displacement after 2 seconds is 45 metres to the right of the origin.

- a Calculate the initial velocity of the particle.
- b Find the displacement,  $x(t)$ , of the particle at time  $t$ .
- c Calculate the displacement of the particle after four seconds.
- d Find the total distance travelled between two and four seconds.

- 8 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 4t - 4$$

The particle is instantaneously stationary when it is 1 m right of the origin.

- a Find the time when the particle is stationary.
- b Find the displacement,  $x(t)$ , of the particle at time  $t$ .
- c Find the value of  $x(t)$  at:

i  $t = 0$

ii  $t = 1$

iii  $t = 2$

- 9 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 6t^2 - 14t + 3$$

The displacement after 2 seconds is 4 metres to the left of the origin.

- a Calculate the initial velocity of the particle.
- b Find the displacement,  $x(t)$ , of the particle at time  $t$ .
- c Calculate the displacement of the particle after four seconds.
- d Assuming the object continues to move in the same direction, find the total distance travelled between two and four seconds.

- 10 The velocity  $v(t)$ , in metres per second, of an object travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 12t^2 - 48t, \text{ where } t \geq 0$$

The object is initially 5 m to the right of the origin.

- a Find the displacement,  $x(t)$ , of the particle at time  $t$ .
- b Find the times,  $t$ , when the object is at rest.
- c Find the displacement at which the object is stationary other than its initial position.

- 11 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 3t^{\frac{3}{2}} - 2t$$

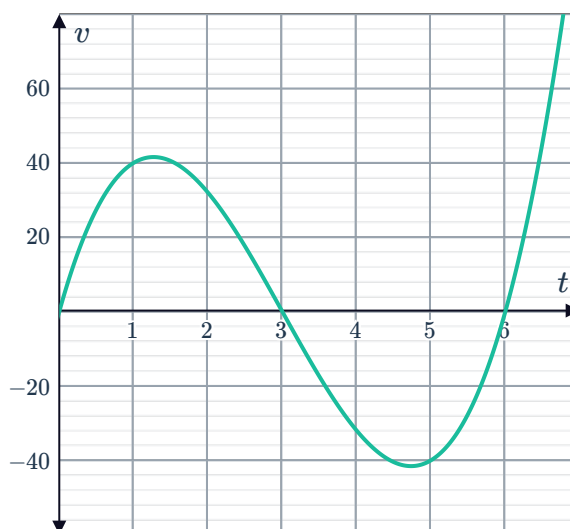
- Find the initial acceleration of the particle.
- If the particle is initially at the origin, find an expression for the displacement  $x(t)$ .
- Find the exact time  $t > 0$  at which the particle is stationary.
- How far does the particle travel in the first 2 seconds? Round your answer correct to two decimal places.

- 12 A pen moves along the  $x$  axis ruling a line. The diagram shows the graph of the velocity of the tip of the pen as a function of time.

The velocity, in centimetres per second, after  $t$  seconds is given by:

$$v(t) = 4t^3 - 36t^2 + 72t$$

When  $t = 0$ , the tip of the pen is at  $x = 5$ .



- Find an expression for  $x(t)$ , the position of the tip of the pen, as a function of time.
  - What feature will the graph of  $x(t)$  have at the point where  $t = 3$  and how does this relate to the movement of the pen?
  - The pen uses 0.02 milligrams of ink per centimetre travelled. Find the amount of ink used between  $t = 0$  and  $t = 4$ .
- 13 The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 5(1 - e^{-t})$$

- Calculate the exact total change in displacement over the interval  $[0, 5]$ .
- Hence, calculate the average change in displacement over the interval  $[0, 5]$ , correct to three decimal places.
- Use technology to calculate the average change in the distance over the interval  $[0, 5]$ , correct to three decimal places.
- For the given velocity function is the average change in displacement equal to the average change in distance for  $t \geq 0$ ? Explain why or why not.

- 14 The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a = 2t - 15, \text{ where } t \geq 0$$

The object is initially moving to the right at 56 m/s.

- a Find the velocity,  $v(t)$ , of the particle at time  $t$ .
  - b Find for all times at which the particle is at rest.
- 15 The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 6t - 27, \text{ where } t \geq 0$$

After 10 seconds, the object is moving at 90 m/s in the positive direction.

- a Find the velocity,  $v(t)$ , of the particle at time  $t$ .
  - b Find all the times at which the particle is at rest.
- 16 The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of an object travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 2, \text{ where } t \geq 0$$

The object is initially 8 m to the right of the origin and moving to the left at 5 m/s.

- a Find the velocity,  $v(t)$ , of the particle at time  $t$ .
  - b Find the displacement,  $x(t)$ , of the particle at time  $t$ .
  - c Find the position of the object at 7 seconds.
  - d Find the time at which the particle is moving at a speed of 3 m/s to the right.
- 17 A coin is thrown vertically upwards from the top of a building 30 m high with an initial velocity of 20 m/s. Using the approximation of acceleration due to gravity of  $-10 \text{ m/s}^2$ , find:
- a The time,  $t$ , taken for the coin to reach its maximum height.
  - b The maximum height reached by the coin.
  - c The time,  $t$ , taken for the coin to reach the ground.
  - d The velocity of the coin as it hits the ground.



- 18** An object is thrown vertically upwards with an initial velocity of  $15 \text{ m/s}$  and an initial height of  $2 \text{ m}$ . Using the approximation of acceleration due to gravity of  $-9.8 \text{ m/s}^2$ , find:
- a** The time,  $t$ , taken for the object to reach its maximum height.
  - b** The maximum height reached by the object, correct to one decimal place.
  - c** The time,  $t$ , taken for the object to reach the ground, correct to one decimal place.
  - d** The velocity of the object as it hits the ground, correct to one decimal place.

- 19** The acceleration  $a(t)$ , in  $\text{cm/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = -e^{4t}$$

The particle is initially at rest.

- a** Find the velocity,  $v(t)$ , of the particle at time  $t$ .
  - b** Find the change in the particle's position after 5 seconds, to the nearest centimetre.
- 20** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = e^{-t}$$

The particle starts off from the origin with an initial velocity of  $2 \text{ m/s}$ .

- a** Find the velocity of the particle after 4 seconds, correct to two decimal places.
  - b** Find the displacement of the particle after 4 seconds, correct to the nearest hundredth of a metre.
- 21** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 2e^{3t}$$

The initial velocity of a particle is  $10 \text{ m/s}$ .

- a** Find its velocity,  $v(t)$ , after  $t = 2$  seconds. Round your answer to the nearest metre per second.
- b** Find the displacement of the particle after 4 seconds, given the initial position is  $2 \text{ m}$  to the left of the origin. Round your answer to the nearest metre.



## Trigonometric applications

- 22** The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 6 \sin 3t$$

- a** Find the acceleration of the particle at  $t = \frac{\pi}{3}$ .
- b** Find the displacement of the particle at  $t = \frac{\pi}{9}$ , given that when  $t = 0$ , the displacement is 2 m.

- 23** The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 6 \cos 3t$$

- a** Find the acceleration of the particle at  $t = \frac{\pi}{9}$ .
- b** Find the displacement of the particle at  $t = \frac{\pi}{3}$ , given that when  $t = 0$ , the displacement is 3 m.

- 24** The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 4 \sin \left( \frac{\pi t}{3} \right)$$

- a** Find the change in displacement in the first 8 seconds.
- b** Given that the distance function is the absolute value of the displacement function, calculate the change in distance in the first 4 seconds.

- 25** The velocity  $v(t)$ , in metres per second, of a particle travelling horizontally along a straight line after  $t$  seconds is given by:

$$v(t) = 6 \cos \left( \frac{\pi t}{4} \right)$$

- a** Find the change in displacement in the first 9 seconds.
- b** Find the change in distance in the first 8 seconds.

- 26** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle  travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 64 \sin 4t$$

- a** Find  $v(t)$ , the velocity in  $\text{m/s}$  at time  $t$ , if the particle is initially traveling at  $2 \text{ m/s}$ .
  - b** Hence, calculate the speed of the particle at  $t = \frac{\pi}{3}$ .
  - c** Find  $x(t)$ , the displacement in metres at time  $t$ , if the particle is initially located  $3 \text{ m}$  from the origin.
  - d** Hence, calculate the displacement at  $t = \frac{\pi}{3}$ .
- 27** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 50 \cos 5t$$

- a** Find  $v(t)$ , the velocity in  $\text{m/s}$  at time  $t$ , if the particle is initially traveling at  $6 \text{ m/s}$ .
  - b** Hence, calculate the speed of the particle at  $t = \frac{\pi}{6}$ .
  - c** Find  $x(t)$ , the displacement in metres at time  $t$ , if the particle is initially located at  $3 \text{ m}$  from the origin.
  - d** Hence, calculate the displacement of the particle at  $t = \frac{\pi}{6}$ .
- 28** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 6 \sin \left( \frac{\pi t}{3} \right)$$

- a** Find the change in velocity between  $t = 1$  and  $t = 7$ .
  - b** Given that the speed function is the absolute value of the velocity function, calculate the total change in speed in the initial 4 seconds.
- 29** The acceleration  $a(t)$ , in  $\text{m/s}^2$ , of a particle travelling horizontally along a straight line after  $t$  seconds is modelled by:

$$a(t) = 4 \cos \left( \frac{\pi t}{4} \right)$$

- a** Find the change in velocity between  $t = 1$  and  $t = 3$ .
- b** Calculate the total change in speed in the first 4 seconds.